Oligopoly

market structure

"few" firms: firms have to behave "strategically"
- i.e. take into account other's behaviour

Cournot Model

Firms strategically choose quantities to produce (identical goods)
→ market price depends on total quantity

ex. two firms (duopoly) in a market produce identical goods

demand curve: $P = 100 - Q$ where $Q = q_1 + q_2$ individual firm quantities

each firm has total cost: $TC_1 = 10q_1$, $TC_2 = 10q_2$ } each has $MC = 10$

thought experiment:

suppose firm 2 produced $q_2 = 0$
→ what would firm 1's optimal quantity be? (best response)
→ here, firm 1 would effectively be a monopolist

demand: $P = 100 - q_1 - 0$ $\quad MC = 10$

$MR = 100 - 2q_1$

$MR = MC \Rightarrow q_1 = 45$ ⟶ firm 1's best response to $q_2 = 0$

how should firm 1 respond if firm 2 produces $q_2 = 20$?

demand: $P = 100 - q_1 - 20 = 80 - q_1$

$MR = 80 - 2q_1$

$MC = 10$

$MR = MC \Rightarrow q_1 = 35$

note: Firm 2's output decision affects firm 1's optimal choice

- as firm 2 produces more output, firm 1's best response is to reduce output

<u>Cournot Equilibrium</u>: each firm's output q_1^* & q_2^* is a best response to the other firm's output

- no "regrets"
- Nash Equilibrium

<u>Find the Cournot Equilibrium</u>

Step 1) Find the best responses for both firms in general → best response functions

→ suppose firm 2 produces q_2, how should firm 1 respond?

Demand: $P = 100 - q_1 - q_2$

Firm 1's total revenue: $TR = P \cdot q_1$

$$TR = (100 - q_1 - q_2) \cdot q_1$$
$$TR = 100q_1 - q_1^2 - q_1 q_2$$

So, Firm 1's marginal revenue is: $MR = 100 - 2q_1 - q_2$

$$MR = MC \Rightarrow 100 - 2q_1 - q_2 = 10$$
$$q_1^* = \frac{90 - q_2}{2} \longrightarrow \text{best response function for Firm 1}$$

the same function would result for Firm 2 here (both have identical costs)

$$q_2^* = \frac{90 - q_1}{2} \longrightarrow \text{best response function for Firm 2}$$

Cournot Equilibrium: both firms choose best responses

→ solve these together

$$q_1 = \frac{90 - q_2}{2} \rightarrow \text{sub in } q_2^*$$

$$q_1 = \frac{1}{2}\left(90 - \frac{90 - q_1}{2}\right)$$

$$2q_1 = 90 - \frac{90-q_1}{2}$$

$$4q_1 = 180 - 90 + q_1$$

$$3q_1 = 90$$

$$\left.\begin{aligned} q_1^* &= 30 \\ q_2^* &= \frac{90-30}{2} = 30 \end{aligned}\right\} \text{Cournot Equilibrium Quantities}$$

Market price is $P = 100 - q_1 - q_2$

$P = 100 - 30 - 30$

$P = 40$

How much profit does each firm make in Cournot Equilibrium? (Remember $TC = 10q$)

$$\pi_1 = P \cdot q_1 - TC = 40 \cdot 30 - 10 \cdot 30 = 900$$

$$\pi_2 = P \cdot q_2 - TC = 40 \cdot 30 - 10 \cdot 30 = 900$$

Question: what if these firms formed a cartel?

→ stop competing with each other and choose the market quantity cooperatively?

How would they choose Q? ← the monopoly quantity

$P = 100 - Q$ $\qquad$ $MR = 100 - 2Q$

$TR = PQ$ $\qquad$ $MR = MC \Rightarrow 100 - 2Q = 10$

$= (100 - Q)Q$ $\qquad$ $Q = 45$

$= 100Q - Q^2$ $\qquad$ $P = 100 - 45 = 55$

How much do the firms make in this cartel?

Total Cartel profit: $\pi = PQ - 10Q = (55)(45) - 10(45) = 2025$

If firms split evenly → each gets \$1012.50

<u>Cartel</u>: Firms reduce output

- market quantity $Q\downarrow$
- market price $P\uparrow$
- firm profits $\pi\uparrow$

would the cartel survive?

- suppose firm 2 sticks to the cartel agreement and produces $q_2 = 22.5$ (half the cartel output)
- could firm 1 make even more money by cheating on the cartel agreement?
- firm 1 would best respond to $q_2 = 22.5$

$$q_1 = \frac{90-22.5}{2} = 33.75$$

now price becomes $P = 100 - 33.75 - 22.5$
$= 43.75$

profits: $\pi_1 = (43.75)(33.75) - (10)(33.75) = 1139$
$\pi_2 = (43.75)(22.5) - (10)(22.5) = 759$